Updating and Expanding Inferred Disease and Phenotype Annotations at Mouse Genome Informatics

Goals:

1. Refine and expand the rules for inference of Gene to Human Disease and Gene to Phenotype annotations from existing Genotype to Disease and Genotype to Phenotype annotations
2. Develop rules for inferring Allele to Human Disease and Allele to Phenotype annotations from Genotype to Human Disease and Genotype to Phenotype annotations

Methods:

1. Develop ruled based algorithms to identify genotypes with a single causative gene or allele
2. Rules rely on meta-data annotations of
    1. Genotypes
    2. Alleles
    3. Genome features
3. Rules
    1. Eliminate non-causative alleles
    2. Count genome features associated with allele
    3. Eliminate genotypes with more than one associated gene or allele except in few special cases
    4. Create new inferred gene or allele-level annotations from remaining genotypes

Meta-data:

1. Genotype
    1. Conditional flag
2. Allele
    1. Attributes
        1. Inserted expressed sequence (IES)
        2. Recombinase
        3. Reporter
        4. Transactivator
    2. Genome feature associations
        1. Allele of – endogenous or transgenic genome feature of the allele
        2. Expressed components – any exogenous gene expressed by the allele
        3. Mutation involves – all endogenous genome features mutated by the allele, including genes targeted by iRNA constructs
    3. Wild-type
    4. Generation method
        1. Transgenic
3. Genome feature
    1. Type
        1. Heritable phenotype marker
        2. Complex/cluster/region
        3. Cytogenetic marker and descendants
    2. Docking sites –
        1. Genes commonly used to insert exogenous sequence without disrupting the endogenous gene or where the endogenous gene is not associated with any phenotype
            1. *Col1a1*
            2. *Hprt*
            3. *Gt(ROSA)26Sor*

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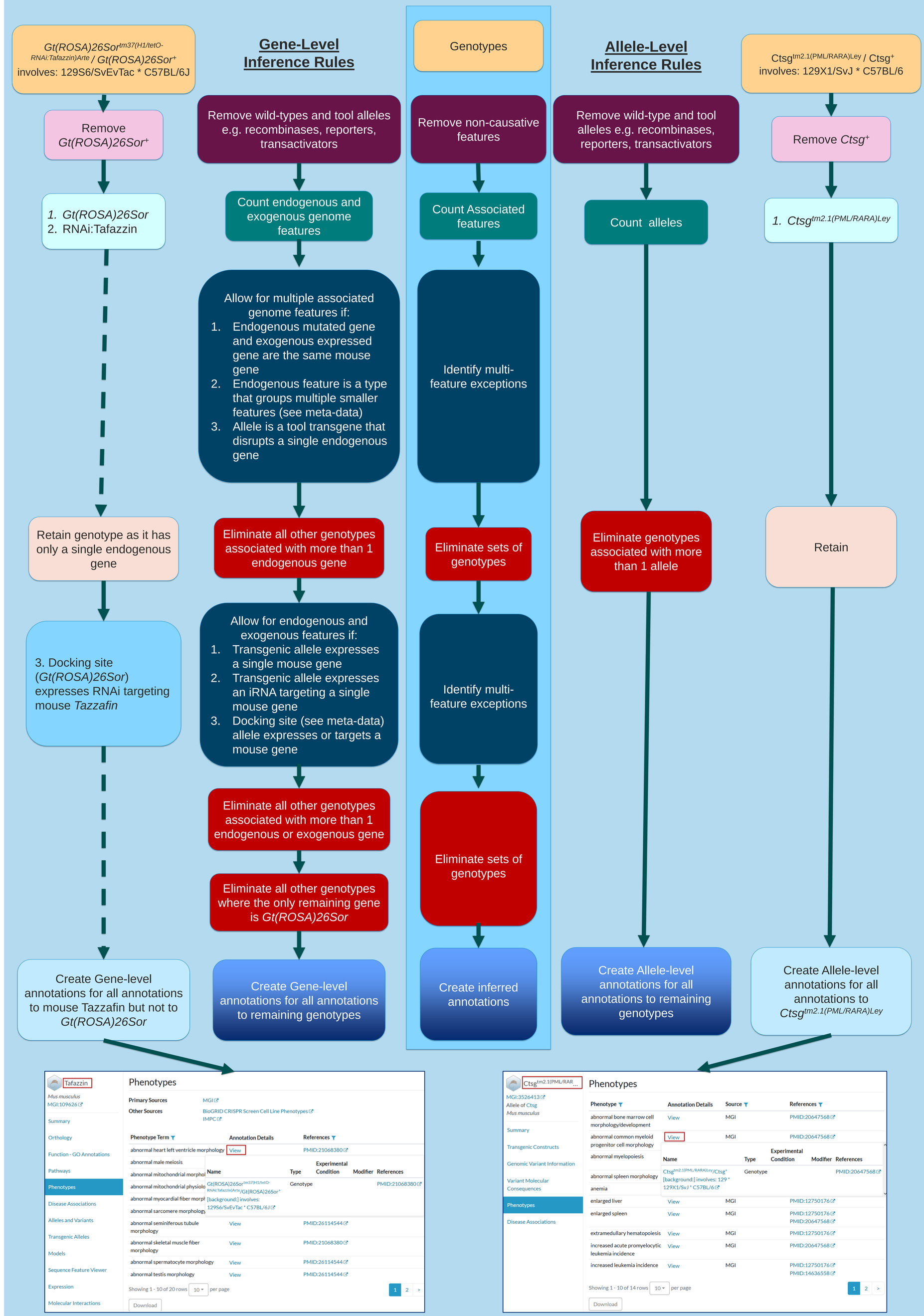

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